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Preictal period optimization for deep learning-based epileptic seizure prediction

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Abstract

Objective. Accurate seizure prediction could prove critical for improving patient safety and quality of life in drug-resistant epilepsy. While deep learning-based approaches have shown promising performance using scalp electroencephalogram (EEG) signals, the incomplete understanding and variability of the preictal state imposes challenges in identifying the optimal preictal period (OPP) for labeling the EEG segments. This study introduces novel measures to capture model behavior under different preictal definitions and proposes a data-centric deep learning methodology to identify the OPP. *Approach*. We trained a competent subject-specific CNN-Transformer model to detect preictal EEG segments using the open-access CHB-MIT dataset. To capture the temporal dynamics of the model's predictions, we fitted a sigmoidal curve to the model outputs obtained from uninterrupted multi-hour EEG recordings prior to seizure onset. From this fitted curve, we derived key performance measures reflecting the timing of predictions, including classifier convergence, average error, output stability, and the transition between interictal and preictal states. These measures were then combined to compute the Continuous Input–Output Performance Ratio, a novel metric designed to comprehensively compare model behavior across different preictal definitions (60, 45, 30, and 15 min) and suggest the OPP for each patient. *Main results*. The CNN-Transformer model achieved state-of-the-art performance (area under the curve of 99.35% and $F1$-score of 97.46%) using minimally pre-processed EEG signals. The 60-minute preictal definition was associated with earlier seizure prediction, lower error in the preictal state, and reduced output fluctuations, leading to significantly higher CIOPR scores ($p < 0.001$). Conventional accuracy-related metrics (sensitivity, specificity, F1-score) were less sensitive to varying preictal definitions and often discordant with CIOPR findings. Cross- and intra-patient heterogeneities in the prediction times were also observed, complicating the establishment of a global preictal interval. *Significance*. The newly developed metrics demonstrate that varying the preictal period significantly impacts the timing of predictions in ways not captured by conventional accuracy-related metrics. Understanding this impact and the inter-seizure heterogeneities is essential for developing intelligent systems tailored to individual patient needs and for underlining practical limitations in detecting the preictal period in real-world clinical applications.

1. Introduction

Epilepsy stands as one of the most prevalent neurological conditions worldwide, impacting an estimated 65 million individuals. This disorder is defined by recurrent epileptic seizures–unprovoked surges of electrical activity in the brain, leading to transient alterations in behavior or consciousness. Patients span diverse socio-demographic backgrounds, with a lifetime prevalence rate of 3%. Notably, an approximate of 30% of patients suffer from drug-resistant epilepsy [1].

An inherent risk factor contributing to mortality among epileptic patients is the unpredictable nature of seizure occurrences, leading to elevated levels of uncertainty, anxiety, social stigma, distress, and potentially hazardous situations among patients [2]. These challenges not only diminish their quality of life but also underscore the critical need for effective seizure management strategies.

Improvements in predicting epileptic seizures would significantly reduce the burden of uncertainty for patients, enabling them, along with caregivers and clinicians, to take preparatory actions, prevent injuries, and perform timely interventions [3, 4]. Epileptic seizures could be predicted by identifying the preictal state, a period preceding seizure onset (ictal state) characterized by distinct morphological EEG differences from the interictal (normal) state. It can typically last from several minutes to a few hours, varying upon seizures and individuals [5].

1.1. Deep learning for epileptic seizure prediction

Deep learning algorithms have gained increasing popularity in epileptic seizure prediction owing to their ability to learn complex features from data [4]. Convolutional neural networks (CNN) have demonstrated particular efficacy in automatically extracting underlying patterns of preictal activity from raw EEG signals [6], EEG wavelets [7], time-frequency data matrices [8], and connectivity-based measures [9]. Recurrent neural networks (RNN), including gated recurrent units (GRU), long short-term memory (LSTM), and bi-directional LSTM (BiLSTM), have been successfully employed to capture temporal dependencies in preictal EEG feature vectors [10, 11]. However, when applied to raw EEG input, RNNs exhibit inferior performance compared to CNNs [12]. Various studies [6, 12, 13] have proposed hybrid models combining the advantages of both architectures with superior overall performance. While direct comparison of reported performances across different studies is not straightforward due to differences in experimental setups, ablation studies within a given work can showcase performance improvements due to specific design choices. Authors in [6] demonstrated that adding a Bi-LSTM layer to a CNN model can improve the accuracy of a subject-specific classifier from 94.1% to 99.7%. Similarly, ablation analysis in a cross-patient setting [13] showed that adding CNN layers prior to a GRU classifier led to a 4% increase in accuracy, illustrating the superiority of hybrid models.

Attention mechanisms, especially within transformer models, have improved the ability to learn data relationships irrespective of sequence distance, a significant limitation in traditional LSTM applications [14, 15]. Integrating CNNs with transformers has led to the emergence of robust models that effectively predict seizures from both raw EEG [16–18], EEG connectivity maps [19], and frequency-domain features [18, 20, 21]. Other studies have successfully combined LSTM layers with transformer models and achieved state-of-the-art performance [22, 23], or employed alternative attention mechanisms, such as criss-cross-attention and channel attention [24].

1.2. Determining the preictal period length

Despite these technological advances, accurately identifying the optimal preictal period (OPP) [5] for labeling EEG segments remains a formidable challenge. This difficulty stems from the gradual nature of the transition into the preictal state and the significant intra- and inter-patient variability in ictogenesis [25, 26]. Approaches to determining the OPP are typically categorized into pre-training and post-training. The former involves defining the preictal length for each seizure before model training through the analysis of EEG markers. Researchers in [5] defined the OPP as the point at which the discriminability of spectral features between the two classes was maximized, which on average occurred 44.3 min before the onset. Works [25, 26] have computed the preictal length using unsupervised clustering methods from nonlinear, univariate, multivariate, and connectivity EEG measures. Preictal events were observed on average up to 47.6 ± 27.3 min before onset, with strong inta- and inter-patient variabilities. As for the post-training approaches, the performance of models trained with varying preictal lengths–usually ranging from 5 to 120 min–is evaluated against predefined metrics, such as accuracy, sensitivity, specificity, and $F1$-score, with the most successful model being selected [7, 8, 10, 27].

Although the aforementioned strategies have helped mitigate the uncertainty in defining the preictal period, they still exhibit considerable limitations. Pre-training approaches are constrained by the quality and relevance of the hand-crafted EEG markers, which may not accurately represent the underlying dynamics of preictal activity. Often the extracted features appear indistinguishable among the two states, yielding no clear definition of the preictal state start [5, 26]. Post-training methods, though less sophisticated, demonstrate greater resilience to our incomplete understanding of preictal state dynamics and align more directly with the targeted diagnostic task. Nonetheless, conventional metrics used for model comparison fail to provide a comprehensive assessment of system performance. Though indicative of the model's ability to distinguish preictal from interictal segments, they capture neither the classifier's behavior during the gradual transition to the preictal state nor the distribution of false positives and negatives, which are critical for evaluating the model's practical implementation. Few studies [28–30] have employed continuous, long-term EEG to visualize the system's behavior, yet they still rely on standard metrics without introducing novel measures.

1.3. Contributions

Recognizing these gaps, this study introduces a novel methodology for evaluating epileptic seizure prediction models. By training a high-performing classifier, we aim to provide a comprehensive assessment that not only monitors the model's behavior but also facilitates the selection of the OPP. Therefore, the contributions of this work can be summarized as:

(1) We develop a CNN-transformer model to classify spatiotemporal EEG dynamics of preictal versus interictal EEG segments achieving high sensitivity and early prediction consistently across subjects. CNN-Transformer models have been previously introduced but our architecture achieves–to the best of our knowledge–the earliest prediction time reported in literature (76.8 min before onset) while maintaining accuracy comparable to state-of-the-art studies.

(2) We introduce an approach that allows nuance assessment of the deep learning model's performance by fitting a sigmoidal curve to the output of the classifier subject to continuous, long EEG input.

(3) We developed a novel Continuous Input–Output Performance Ratio (CIOPR) metric that provides a comprehensive assessment of the performance of a seizure prediction system in a realistic implementation setting by combining measures of prediction time, output stability, and transition time between interictal and preictal states.

(4) We demonstrate the impact of different preictal state definitions in the system's performance as well as how the CIOPR metric can be utilized to determine the OPP for each patient (OPP).

2. Materials and methods

2.1. EEG data and participant selection

The proposed seizure prediction model was trained and evaluated on the CHB-MIT dataset [31]. It consists of scalp EEG recordings from 23 pediatric patients at the Children's Hospital Boston following an anti-seizure medication withdrawal period of seven days. Patients' ages ranged from 1.5 to 22 years, with individual epilepsy types not being mentioned. There were 198 recorded seizures in total, for which seizure start and end times were annotated by experts. Cases *chb01* and *chb21* came from the same subject, with the latter being recorded 1.5 years later. Subject information for case *chb24* was not provided. Electrodes were placed according to the international 10–20 system with the number of channels for most of the patients ranging from 23 to 26. The sampling rate was set to 256 Hz with 16-bit resolution.

Data recorded up to 4 h before and 1 h after each annotated seizure were classified as interictal, to account for the uncertainty in the duration of preictal and postictal effects. The 4-h threshold is widely adopted in literature [6, 20, 22, 32, 33], while the postictal threshold varies across studies. Literature has shown that the postictal state typically lasts up to 30 min [34], therefore we selected a 1 h threshold to ensure a sufficient number of training data for cases with more frequent seizures. Cases *chb08*, *chb12*, *chb13*, *chb15*, and *chb24* did not yield interictal data meeting the defined criteria and were deemed ineligible and excluded from the study.

The maximum duration for the preictal period was set to 60 min. Similarly to the interictal class, data collected up to 1 h after seizure termination were not considered preictal regardless of an incoming seizure. Seizures yielding less than one minute of preictal data were excluded as ineligible. Additionally, cases with fewer than two eligible seizures were also excluded, since there should be at least one seizure for training and one for testing each subject-specific model. Cases *chb01* and *chb21* were treated as separate subjects due to the long time difference between the recordings. In instances where electrode placement varied across recordings, the configuration yielding the greatest amount of data was selected. A summary of all patient data and eligible seizures, as well as their *Patient ID*, *Seizure ID*, and file name, are detailed in the supplementary material tables 1 and 2, respectively.

2.2. Pre-processing

Pre-processing of the EEG signals was kept minimal and was performed on the files meeting the eligibility criteria in section 2.1. Firstly, EEG was re-referenced to the common average, which has been shown to improve the signal-to-noise ratio [35]. A 0.5–45 Hz finite impulse response (FIR) non-causal bandpass filter was automatically designed by the MNE Python package using a Hamming window and order $N = 1690$. The high-pass cutoff at 0.5 Hz served to remove DC offset and low-frequency voltage drifts [36]. The 45 Hz low-pass cutoff was chosen to suppress power line interference without using notch filtering, which can cause significant signal distortions [37]. Additionally, recent studies suggest that interictal SEEG features below 45 Hz are sufficient for localizing the seizure onset zone [38]. This design choice aims to balance minimal noise contamination while retaining relevant gamma-band information, exploring the hypothesis that a 45 Hz cutoff could also be effective in scalp-EEG seizure prediction. Lastly, no channels were removed, resulting in 23 channels for each eligible patient. Pre-processed data were then segmented into 5 s non-overlapping epochs.

2.3. Training methodology

Four different preictal period lengths – 60, 45, 30, and 15 min–were then extracted from each seizure to explore their effect on model performance and determine the OPP for each subject separately. These intervals were selected to ensure an adequate number of seizures were available for training the subject-specific deep learning models while maintaining a manageable number of comparisons.

Data extraction was conducted as follows: for the 60 min duration, all available preictal data were extracted for each seizure, regardless of the actual duration available (e.g. some seizures had only 20 min of data available). For the 45 min duration, only segments from the 45 min immediately preceding the seizure onset were included; if a seizure had 60 min of preictal data available, the first 15 min were excluded. If, for example, only 20 min were available, none were excluded. This procedure was similarly applied to the 30 min and 15 min durations. A histogram of available preictal data per seizure (in minutes) can be found in the supplementary material, figure 1.

The models were trained and evaluated in a subject-specific manner. For each definition of preictal length, the extracted preictal segments from all seizures were concatenated without shuffling. To mitigate class imbalance, the minority class (preictal) data were oversampled by a factor of 3 through the introduction of a 66% overlap. Similarly, interictal segments were randomly selected from the pool of interictal data to match the number of the augmented preictal data instances. The training and testing sets were created using the leave-one-seizure-out cross-validation (LOOCV) method. Specifically, preictal data of a different seizure each time were isolated, alongside an equal number of randomly selected interictal segments to form the testing set. The remaining samples were shuffled and divided into a 90% training and 10% validation split, forming the training set. This process was repeated four times to ensure consistency across different shuffling and random weight initializations. Ultimately, this resulted in $N \times 4 \times k \times 4$ training and testing sets, where N represents the number of eligible subjects, k denotes the number of eligible seizures per patient, and the four iterations correspond to the different preictal lengths (60, 45, 30, and 15 min), as well as the number of different runs.

2.4. Deep learning model

Following the accelerating emergence of transformer models in seizure-related applications, we decided to employ a hybrid CNN-Transformer deep learning model for the classification of preictal and interictal EEG segments. The model architecture is illustrated in figure 1. The raw pre-processed EEG epochs were first input to a three-layer CNN to extract relevant spatiotemporal features. Each convolutional layer, consisting of 32, 64, and 128 filters of size 3×3, respectively, was followed by a Max-Pooling operation that reduced dimensionality by a factor of two. The architecture also incorporated a Dropout layer with a probability of $p = 0.1$ before each Max-Pooling step to mitigate overfitting, complemented by batch normalization and a rectified linear unit (ReLU) activation function.

After the final convolutional layer, the output tensor was reshaped into a two-dimensional matrix, where rows corresponded to subsequent time points and columns to spatiotemporal features, including positional information resulting from integrating the channel and feature dimensions during the reshaping process. To capture long-term temporal dependencies among these features, a transformer architecture was utilized, with two multi-head attention layers [14]. The reshaped two-dimensional output was passed through 3 linear layers per attention head to create the query, key, and value matrices, respectively. The linear layers shrunk the dimension of the feature vectors to 64, while the number of attention heads was set to 8, following experimentation and literature suggestions [39]. Each attention layer was followed by a fully-connected (dense) layer of 64 neurons, ReLU activation and $p = 0.3$ dropout, to highlight the most relevant features, while maintaining the dimensionality of the feature vectors.

The resulting attention-weighted feature map was then flattened and processed through a single-neuron output layer with sigmoid activation for the classification stage. The output value was rounded to the nearest integer (0: interictal; 1: preictal) without further post-processing. The model was trained using the binary cross-entropy loss function [40], with weight updates performed via the Adam optimization algorithm. The learning rate was set to $l = 0.001$, and the AMSGrad [41] extension was enabled. The

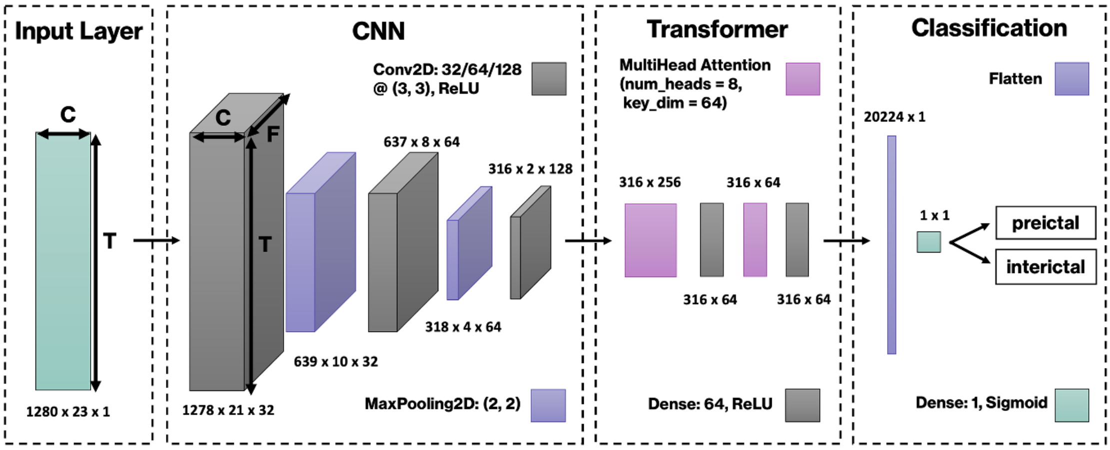


Figure 1. Architecture of the CNN-Transformer deep learning model. Inputs were of shape 1280 × 23, for 5 s 23-channel EEG segments. Dimensions T, C, and F correspond to time points, channels, and feature maps respectively. The shape of the data following each layer is displayed as $T \times C \times F$. For the transformer stage, C and F are aggregated into a single dimension and expressed as $T \times (C, F)$. Specific information on each layer can be found in the supplementary material, table 3.

training utilized a mini-batch size of 64 and was limited to 100 epochs. To prevent overfitting, the early-stopping callback terminated training if the validation loss did not improve for 20 consecutive epochs. The model-checkpoint callback preserved the weights of the training iteration that achieved the lowest validation loss. The complete summary of the model can be found in the supplementary material.

2.5. CIOPR

We introduce a novel metric that facilitates a direct comparison of classifier behaviors across prediction tasks. Traditional accuracy metrics, which rely heavily on class definitions, often fail to account for prediction timing and performance during interictal-preictal state transitions. To address these limitations, our innovative metric, CIOPR, offers a comprehensive evaluation of model behavior using uninterrupted, unlabeled long-term EEG data, allowing for objective comparisons independent of class labels. We hypothesized that the ideal classifier would provide early prediction, minimal errors, stable outputs, and brief transition periods (TPs). Longer transitions can lead to prolonged fluctuations and increased uncertainty in predictions. Therefore, we assumed that shorter transitions are crucial for improving reliability in alarm-generating systems, especially in critical applications like seizure prediction.

When subjected to continuous EEG data spanning several hours before a seizure, an accurate classifier is anticipated to initially produce negative (interictal) predictions, transition through a mix of negative and positive predictions, and ultimately converge to the preictal state. Consequently, to quantitatively model the timing of these predictions, we propose to fit a sigmoidal curve to the classifier's continuous output, which has undergone smoothing with an 8 min averaging window. In particular, a 4-parameter logistic curve is used, given by equation (1) [42], where parameters a, b, c, and d represent the vertical and horizontal stretch, the point of inflection (Infl), and the vertical offset, respectively, as

$$f(x) = \frac{a}{1+\mathrm{e}^{-b(x-c)}} + d. \tag{1}$$

Utilizing the fitted sigmoidal curve, we derive key performance measures. Specifically, the TP between interictal and preictal states is quantified as the time interval between the 5th and 95th percentile thresholds of the sigmoid curve's amplitude. The negative duration (ND), the supposed interictal period, is calculated as the duration between the first output prediction and the beginning of the TP. The remaining measures are directly computed from the output data, following average smoothing (1 output prediction every 8 min). The seizure prediction convergence (SPC) is the point in time where output predictions reached 99% of the maximum value, in minutes before seizure onset. It represents the period of highest incoming-seizure probability by the model and could reflect the prediction horizon in a realistic implementation setting. Both output and maximum values used for the computation of SPC refer to the mean of 3 consecutive predictions (3×8 min $=$ 24 min). Figure 2 depicts the fitted sigmoid, the output predictions, and the aforementioned measures.

The average error during the SPC ($\mathrm{SPC_{err}}$) and ND ($\mathrm{ND_{err}}$) are calculated using equations (2) and (3), respectively. N_{SPC} and N_{ND} refer to the number of output predictions in each of the two regions. $\mathrm{SPC_{err}}$ and $\mathrm{ND_{err}}$ also serve as proxies for output fluctuations, which would cause their values to increase. Then, equation (4) is developed to combine all the measures so that the classifier with the longest SPC, minimum $\mathrm{SPC_{err}}$ and $\mathrm{ND_{err}}$, and shortest TP

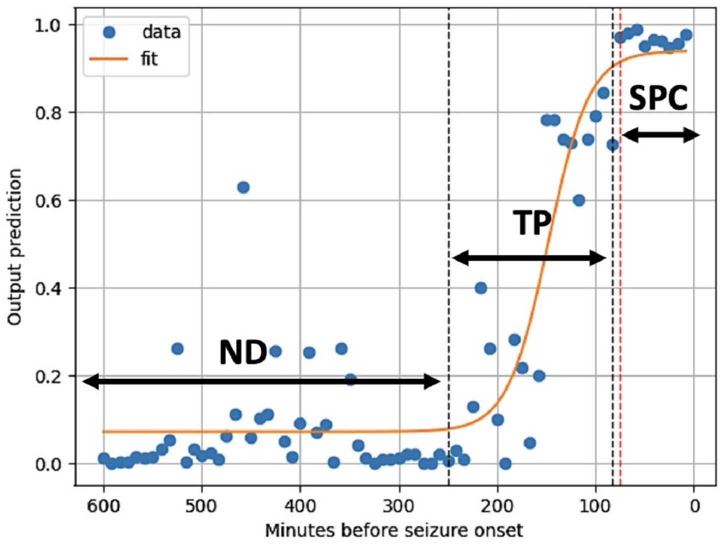


Figure 2. Continuous output predictions based on 600 min of EEG data preceding seizure onset, along with the fitted sigmoidal curve. The black dotted vertical lines, derived from the sigmoid fit, denote the start and end points of the TP between interictal and preictal states. The red dotted line, directly calculated from the output, designates the start of the SPC, where the classifier's output converges. Notations: ND = negative duration (interictal predictions), TP = transition period, and SPC = seizure prediction convergence.

achieved the highest score. SPC_{eff} and ND_{eff} represent the effective values of SPC and ND taking into account the average errors, and defined as $SPC_{eff} = SPC(1 - SPC_{err})$ and $ND_{eff} = ND(1 - ND_{err})$

$$SPC_{err} = \frac{1}{N_{SPC}} \sum_{i=1}^{N_{SPC}} |1 - y_i| \quad (2)$$

$$ND_{err} = \frac{1}{N_{ND}} \sum_{i=1}^{N_{ND}} |y_i| \quad (3)$$

$$CIOPC = SPC_{eff} + \eta \left(ND_{eff} + Infl_{comp}\right). \quad (4)$$

Overall, equation (4) computes the Continuous Input–Output Performance Coefficient (CIOPC) of a model by taking into account two terms. The SPC_{eff} term rewards the model for early prediction (SPC), high accuracy, and output stability (SPC_{err}). Similarly, the second term, ND_{eff}, rewards the model for high accuracy and stability (ND_{err}), as well as short TP (longer ND when TP is short). A reduction in ND due to earlier prediction should not get penalized and is compensated by the $Infl_{comp}$ term (ND_{eff} lost due to an earlier Infl). The scaling factor η ensures that the second term will not have a greater weighting than SPC_{eff} due to hours-long interictal EEG; hence, maintaining the focus on early and accurate seizure prediction.

The CIOPC value for each preictal state definition is normalized to the CIOPR metric using equation (5), where $CIOPC_{max}$ denotes the maximal CIOPC obtained across the examined preictal durations. This normalization facilitates a relative performance evaluation, where the highest-scoring model is assigned a CIOPR score of 1, thus enabling a comparative analysis of the impact of varying preictal period lengths. Detailed expressions for $Infl_{comp}$ and η, as well as elucidation of the underlying intuition for CIOPR, are detailed in the supplementary material, section 3

$$CIOPR_k = \frac{CIOPR_k}{CIOPC_{max}}, k \in \{60, 45, 30, 15\}. \quad (5)$$

Lastly, the Pearson correlation coefficient, ρ is calculated between the output predictions and the fitted curve to assess the method's reliability. It also serves as an indicator of output stability, since greater fluctuations are expected to decrease the correlation coefficient.

2.6. Additional metrics

Beyond the CIOPR performance metrics including segment-wise sensitivity, specificity, accuracy, and $F1$-score were employed for performance evaluation. These metrics–referred to hereafter as conventional metrics due to their widespread use in classification tasks–were computed on the test set, the formulation of which is detailed in section 2.3. Given the equal representation of classes in the test set, the resulting accuracy is equivalent to the balanced accuracy (BA).

2.7. Testing setup

All parameters in the chosen CNN-Transformer architecture, including the decision boundary of

$p = 0.5$ were kept identical across all patients and testing seizures. For each eligible seizure of a qualifying participant (criteria detailed in section 2.1), the prediction performance was tested using the conventional metrics, for all preictal durations. Seizures with over 2.5 h of uninterrupted continuous EEG data were further subjected to CIOPR evaluation for each preictal interval. A maximum of 10 h of continuous EEG was utilized for CIOPR to reduce computation time and maintain consistency across patients.

Metrics were then aggregated across seizures to compute the average performance for each preictal duration. For conventional metrics, averages were calculated on a segment-wise basis rather than seizure-wise, ensuring that testing seizures with more data samples were accorded greater weight. These averages were then used to determine the OPP. Table 1 of the supplementary material provides a summary of the recorded seizures per subject, including both the count of eligible seizures and those applicable for CIOPR testing.

2.8. OPP selection

A general OPP – 60, 45, 30, or 15 min–was then selected for each participant. When at least one seizure was eligible for CIOPR testing, the OPP was assigned to the preictal definition yielding the highest CIOPR values, averaged across those seizures. Conversely, if no CIOPR results were available, the OPP was assigned to the preictal definition attaining the highest $F1$-score, averaged across all testing seizures. $F1$-score is preferred over BA due to its capability to emphasize minimizing false negatives, which are considered more detrimental in epileptic seizure prediction [4]. Lastly, the model trained using the OPP was selected from each patient to evaluate both the subject-specific and aggregate performance of our proposed methodology. To further showcase performance, the false alarm rate (FAR) (h^{-1}) and the area under the receiver operating characteristic (ROC) curve (AUC) were reported only for the selected models. FAR was computed based on the EPOCH [43] method on the 5 s segment level.

2.9. Statistical analysis

To explore statistical differences in model behavior attributed to different preictal definitions, a related-samples Friedman's two-way ANOVA with Bonferroni correction for multiple comparisons was conducted for all the metrics used in the testing setup (sensitivity, specificity, accuracy, $F1$-score, SPC, SPC_{err}, ND_{err}, Infl, TP, ρ, CIOPR). The null hypothesis tested was that the distributions of scores for each metric across the preictal lengths of 60, 45, 30, and 15 min do not differ significantly (significance level $\alpha = 0.05$ after correction).

3. Results

3.1. Classifier training

The models were trained using an A100 Tensor Core GPU in Google Colab. The average training duration was 8.1 min per model, converging at epoch 54 due to the early-stopping callback. The training duration was notably longer for 60- and 45 min preictal class definitions due to increased data volume. The convergence minima for both validation and training curves varied more across patients than across intra-patient seizures or different preictal state lengths.

3.2. Preictal-interictal classification

The proposed architecture demonstrated consistent performance across multiple subjects with particularly high sensitivity ($>99\%$) for all preictal definitions. Figure 3 compares the distributions of the achieved sensitivity, specificity, and $F1$-score across all subjects for each definition of the preictal class. The exact values for each patient are detailed in supplementary material, section 4.

Models trained with a 30 min preictal definition demonstrated the highest sensitivity with an average of 99.49%. However, varying the preictal window had minimal impact on sensitivity, which was confirmed by the statistical analysis that showed no significant differences across different preictal definitions (null hypothesis accepted, $p = 0.161$). On the contrary, the analysis showed a more pronounced impact on model specificity, with the null hypothesis being rejected for the 60-to-15 ($p = 0.001$) and 45-to-15 min ($p = 0.034$) preictal length comparisons. Overall, specificity improved with longer preictal intervals and peaked at 95.03%, which was considerably lower than sensitivity at the 0.5 decision threshold.

Furthermore, the 60 min preictal window generally yielded the highest average $F1$-score (97.28%), although 45- and 30 min definitions performed better in certain individuals. Statistical analysis showed significant impact on $F1$-score only for the 60-to-15 min comparison ($p = 0.003$). Overall, the percentage changes in $F1$-score across different preictal intervals were relatively modest, with a maximum variation of 3.2% observed in case *chb05*. Given the minimal impact on sensitivity, fluctuations in $F1$-score were primarily attributable to reductions in specificity associated with the shorter preictal intervals.

Some cases in our study had much lower performance compared to others. For instance, case *chb06* exhibited a sensitivity of 94.76%, and a specificity of 73.54%, $> 20\%$ below the average. Cases *chb14* and *chb22* also showed notably low specificity scores. Section 4.1 compares the findings with the results reported in the literature.

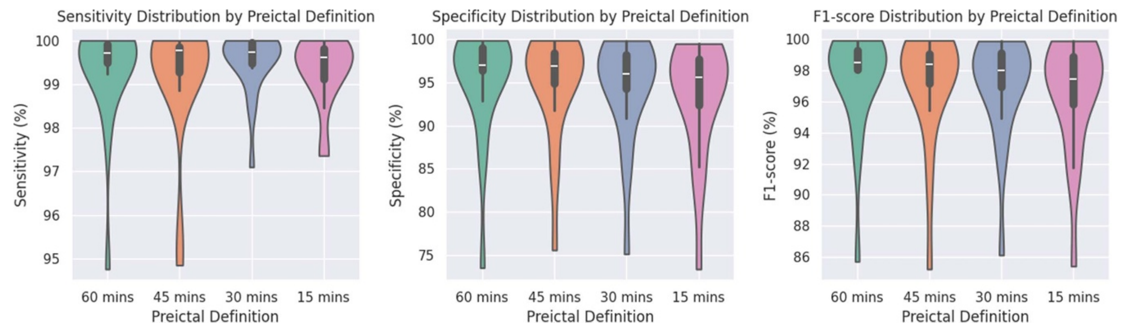
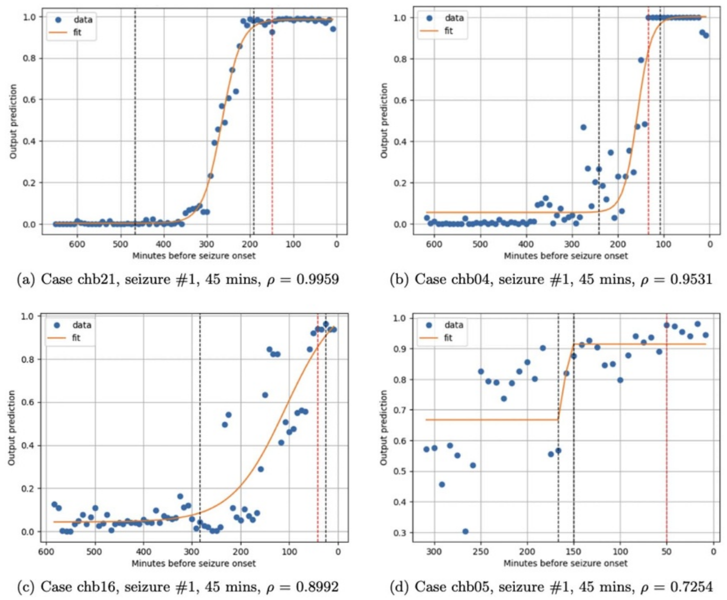


Figure 3. Sensitivity, specificity, and $F1$-score distributions across the 19 eligible subjects of the CHB-MIT dataset for the 60, 45, 30, and 15 min preictal definitions. Significant differences ($p \leqslant 0.05$) were observed across the varying preictal intervals in the following metrics: Specificity (60-to-15, 45-to-15), $F1$-score (60-to-15).



(a) Case chb21, seizure #1, 45 mins, $\rho = 0.9959$ (b) Case chb04, seizure #1, 45 mins, $\rho = 0.9531$ (c) Case chb16, seizure #1, 45 mins, $\rho = 0.8992$ (d) Case chb05, seizure #1, 45 mins, $\rho = 0.7254$

Figure 4. Four figures showing different levels of fitting between the classifier's output and the fitted sigmoid curve, sorted with decreasing Pearson correlation coefficient (ρ). The horizontal axis represents minutes before seizure onset (seizure onset at utmost right), and the vertical axis is the classification output. Captions show the case number, seizure ID, preictal class definition, and ρ.

3.3. Fitting the sigmoidal curve to the classifiers' output

The proposed CIOPR test in section 2.5 was performed on the thirteen patients that met the criteria described in section 2.7. The first step of the analysis involved fitting the sigmoidal curve to the smoothed output of the classifiers. The Pearson correlation coefficient between the model output and the fitted curve across all the tested patients was on average $\rho > 0.9$ for the 60, 45, and 30 min preictal definitions, and $\rho = 0.876$ for the 15 min one. The high correlation values show the effectiveness of the chosen curve in modeling the classifiers' output profile and indicate that the models generally exhibited the intended behavior. Similar to the $F1$-score results, the correlation coefficients differed across patients, seizures and preictal state lengths, with 60 min usually leading to higher values. Statistical analysis showed that increasing the preictal length led to significant increases in the ρ values for the 60-to-15 ($p = 0.005$) and 45-to-15 min ($p = 0.005$) comparisons.

Figure 4 demonstrates four fitting examples to allow visualization of different correlation coefficients and corresponding model behavior. Sub-figures 4(a) and (b) display high correlation coefficients ($\rho > 0.95$). Sub-figure 4(c) depicts a fitting with $\rho \approx 0.9$, enabling reliable identification of both the prediction horizon and the TP. Conversely, sub-figure 4(d) shows one of the fittings with the lowest correlation ($\rho < 0.75$). While the TP and interictal state were less discernible, a noticeable shift in the density of positive predictions still enabled the identification of the predicted preictal state start and subsequent performance quantification. Seizure 2 of case *chb05* was the only instance in which the curve could not be fitted and was therefore excluded from the analysis. The output plots for this case can be found in supplementary material, section 5.

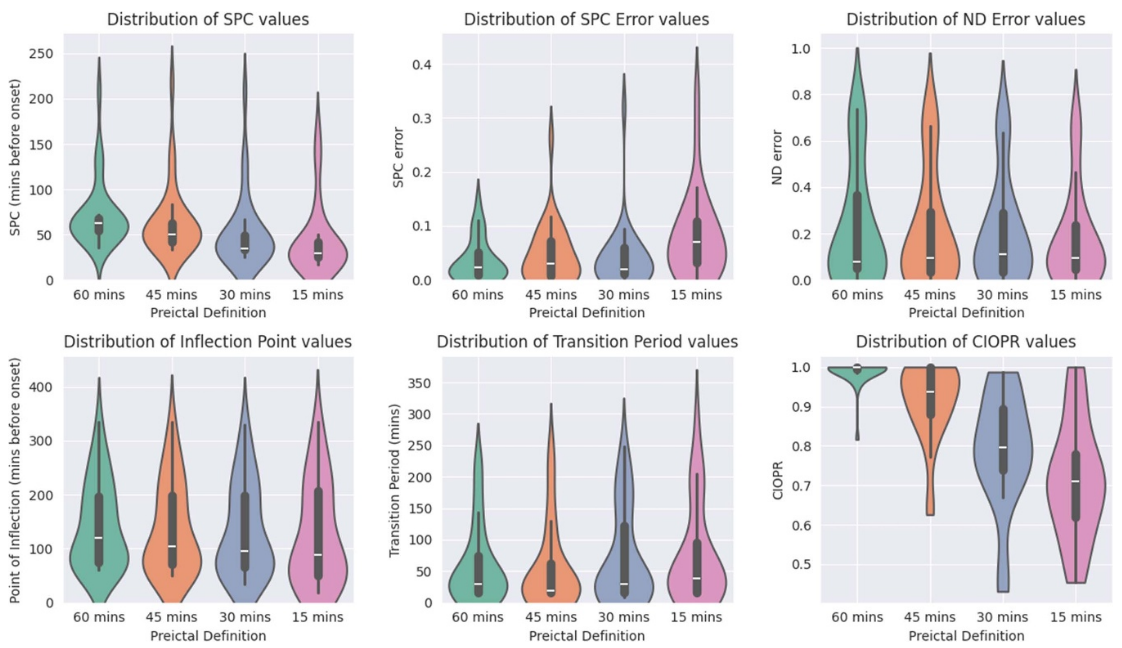


Figure 5. SPC, SPC_{err}, ND_{err}, Infl, TP, and CIOPR distributions across the 13 eligible subjects of the CHB-MIT dataset for the 60, 45, 30, and 15 minpreictal definitions. Significant differences ($p \leqslant 0.05$) were observed across the varying preictal intervals in the following metrics: SPC (60-to-30, 60-to-15, 45-to-30, 45-to-15), SPC_{err} (60-to-45, 60-to-30, 60-to-15), Infl (60-to-30, 60-to-15), CIOPR (60-to-30, 60-to-15, 45-to-30, 45-to-15). Notations: SPC = seizure prediction convergence, SPC_{err} = average error during the SPC, ND_{err} = average error between the start of the signal and TP, TP = transition period, Infl = point of inflection, and CIOPR = continuous input–output performance ratio.

The ρ values were generally reduced with decreasing preictal length, primarily due to the increased volume of 'unseen' data from the classifiers. Since the interictal data were taken up to 4 h before seizure onset irrespective of the preictal window, shorter preictal lengths led to an increased volume of in-between-data that the model had not been exposed to during the training process. This led to prolonged output fluctuations that reduced ρ, as the output diverged from the 'ideal' smooth transition of the sigmoidal curve. A detailed list of average correlation coefficients for all cases can be found in the supplementary material, section 6.

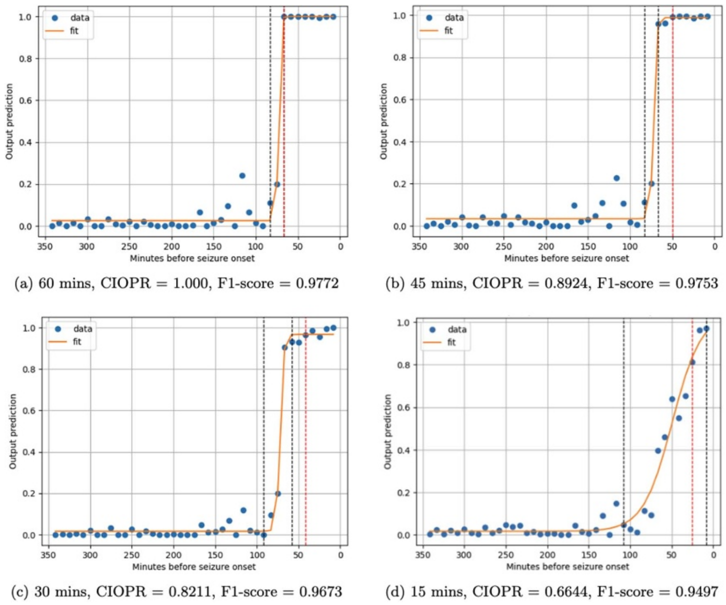
3.4. CIOPR testing results

The fitted curves allowed the computation of the SPC, SPC_{err}, ND_{err}, Infl, TP, and CIOPR metrics, as explained in section 2.5. Figure 5 compares the distribution of the aforementioned metrics across all eligible test patients for each preictal definition. The exact values of all metrics used to compute the CIOPR scores are displayed in the supplementary material, section 6.

Longer preictal definitions led to earlier convergence of the classifier output (SPC), which translates to earlier prediction of incoming seizures. The average convergence point for the 60 min interval occurred at 73.0 min before the onset, while values ranged overall from 217.6 to 16.7 min. Statistical analysis showed that varying preictal definitions had a significant impact on prediction time, particularly for the 60-to-30, 60-to-45, 45-to-30, and 45-to-15 min comparisons. The average error during the SPC (SPC_{err}) appeared to be also significantly impacted by varying preictal windows, with the lowest value observed at the 60 min one. On the other hand, ND_{err} was on average increased for longer preictal intervals. However, this increase was driven by only a few test seizures with high ND_{err} values for all intervals, therefore results for this metric were not deemed statistically significant.

The TP and Infl metrics quantified the timing of the predictions; i.e. the transition from the interictal to the preictal state. Infl occurred earlier when longer preictal intervals, averaging 143.1 min before onset for the 60 min interval. This translates to a tendency to generate positive predictions farther from the seizure onset, with the 60-to-15 and 60-to-30 min comparisons showing statistically significant differences. Interestingly, the transition between the two states appeared sharper for the 60, and 45 min preictal lengths, with the latter achieving the lowest average TP of 56 min. However, no statistical significance on the TP values (null hypothesis accepted with $p = 0.135$) across the different preictal intervals.

The CIOPR values improved with increasing preictal class definitions, with the 60 min preictal window showing the best performance. This was



(a) 60 mins, CIOPR = 1.000, F1-score = 0.9772

(b) 45 mins, CIOPR = 0.8924, F1-score = 0.9753

(c) 30 mins, CIOPR = 0.8211, F1-score = 0.9673

(d) 15 mins, CIOPR = 0.6644, F1-score = 0.9497

Figure 6. Output profiles of four patient-specific classifiers for case *chb02* with 60, 45, 30, and 15 min preictal class definitions respectively. Input is 5.8 h of continuous EEG recordings before the onset of seizure #1 of *chb02*. The red dotted line shows the SPC, and the black lines show the TP boundaries. The horizontal axis represents minutes before seizure onset (seizure onset at utmost right), and the vertical axis is the classification output. Sub-captions show the preictal class definition, CIOPR value, and $F1$-score achieved. Notations: SPC = seizure prediction convergence, TP = transition period, CIOPR = Continuous Input–Output Performance Ratio.

confirmed by statistical analysis, where CIOPR was significantly higher with the 60 min definition compared to the 30 ($p < 0.001$) and 15 min ones ($p < 0.001$). Similar to the $F1$-score results, few individuals showed better CIOPR performance with a 45 min preictal definition (*chb01, chb04, chb22*). The CIOPR scores of the 45 min definition were also significantly higher than the 30 ($p = 0.003$) and 15 min ($p < 0.001$) ones. Detailed p-values across all the hypothesis tests performed can be found in the supplementary material, table 8.

Overall, the CIOPR results were aligned with the SPC trends, due to the design of the metric to positively reward early and consistent predictions. Cases *chb01* and *chb22* were the only exceptions, where they achieved the highest CIOPR values at a 45 min preictal window while having a maximum SPC at the 60 min one. This can be attributed to the integrative nature of the proposed metric, which aims at balancing minimized error, reduced transition time, and early prediction time. Interpretation of the findings and importance in terms of intended behavior in real-time implementation settings is discussed in section 4.3.

3.5. Comparison between CIOPR and $F1$-score metrics

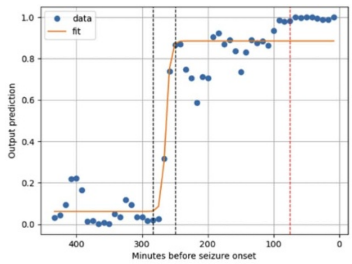
Comparison with the $F1$-score results indicates that CIOPR was considerably more sensitive to varying preictal lengths, showing an average change of 9.2% per 15 min decrease, compared to a 0.3% average change for the $F1$-score. This was also confirmed by the statistical analysis, where the CIOPR distributions varied significantly across 4 preictal length comparisons, compared to only 1 for the $F1$-score, and visual inspection of results. Cases *chb02*, *chb07*, *chb14*, and *chb16* all presented differences ⩾10% between the best and second-best performing models based on the CIOPR metric.

The results can be categorized into two types: a) concordant, where the same preictal length yielded the highest performance in both metrics and b) discordant, where CIOPR and $F1$-score values suggested a different OPP[8]. A detailed comparison between the $F1$-scores and CIOPR can be found in supplementary material, table 6.

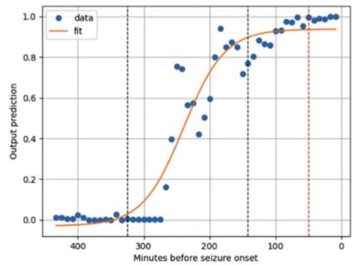
Figure 6 depicts the output curves for case *chb02* alongside the fitted sigmoidal curve. The CIOPR discrepancies can be attributed to the visually identifiable changes in the model's behavior, such as greater prediction horizon, shorter transition time, and reduced error. In this case, the results between the two metrics were concordant; increasing preictal class definition led to improved CIOPR and higher $F1$-score.

On the other hand, cases *chb06*, *chb07*, and *chb14* demonstrated similar CIOPR variations

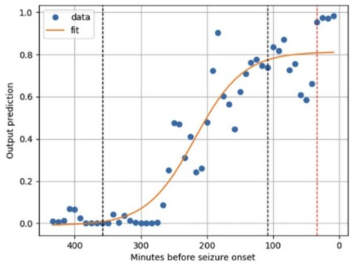
[8] For case *chb21*, the LOOCV approach (see section 2.3) resulted in training seizures with less than 45 min of preictal data when testing for the only eligible seizure (Seizure ID: 1) for CIOPR evaluation. Consequently, the results for the 60 and 45 min preictal periods were identical and the 45 min length was chosen since it provided a more accurate representation of the training instances.



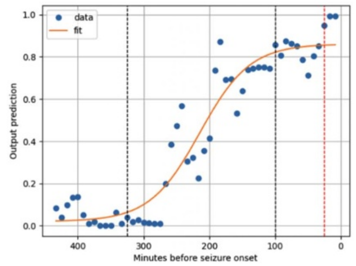
(a) 60 mins, CIOPR = 1.000, F1-score = 0.9838



(b) 45 mins, CIOPR = 0.8893, F1-score = 0.9839



(c) 30 mins, CIOPR = 0.7332, F1-score = 0.9860



(d) 15 mins, CIOPR = 0.7307, F1-score = 0.9777

Figure 7. Output profiles of four patient-specific classifiers for case *chb07* with 60, 45, 30, and 15 min preictal class definitions, respectively. Input is 7.0 h of continuous EEG recordings before the onset of seizure #1 of *chb07*. The red dotted line shows the SPC, and the black lines show the TP boundaries. The horizontal axis represents minutes before seizure onset (seizure onset at utmost right), and the vertical axis is the classification output. Sub-captions show the preictal class definition, CIOPR value, and $F1$-score achieved. Notations: SPC = seizure prediction convergence, TP = transition period, CIOPR = Continuous Input–Output Performance Ratio.

($\geqslant$10%) across different preictal definitions, yet yielded discordant $F1$-score results. In particular, using the example of case *chb07*, the differences in SPC and output stability are depicted in figure 7. Longer preictal periods led to earlier prediction, shorter transition time and reduced output fluctuations, resulting in significantly greater CIOPR. Conversely, the $F1$-scores did not follow the same trend, underscoring the inadequacy of conventional metrics to fully capture the complex temporal behavior of seizure prediction models, as discussed in section 1.2.

Overall, there was a discordance in the best-performing preictal definition between the two metrics in 7 out of the 13 cases that underwent CIOPR testing. This discrepancy occurred only in cases where $F1$-score variations were less than 1% between the two best-performing preictal definitions.

3.6. OPP selection and overall performance

For seizures eligible for CIOPR assessment, the OPP of each patient was determined based on the highest CIOPR values, while for those without CIOPR-eligible seizures, it was based on the highest $F1$-score, as outlined in section 2.7. Overall results are summarized in table 1, which includes the OPP for each patient and corresponding performance metrics. Sensitivity, specificity, FAR, accuracy, AUC, and $F1$-score were computed for the selected OPP using all testing seizures, while SPC used only the ones subject to CIOPR testing. The criterion used to select the OPP (CIOPR or $F1$-score) is also displayed. On average, the subject-specific models achieved a sensitivity of 99.31%, specificity of 95.34%, accuracy of 97.32%, and $F1$-score of 97.46%. FAR averaged 33.6 per hour on a segment-wise basis, since they were computed on the raw model output. However, the average specificity exceeded 95%, suggesting that the introduction of a post-processing scheme would significantly reduce FAR (h^{-1}) in a clinical environment. The classifiers' output converged at 76.8 min before seizure onset, with a standard deviation of 36.8 min, reflecting anticipated cross-patient heterogeneities.

4. Discussion

We trained subject-specific classifiers for 19 pediatric patients of the CHB-MIT dataset to predict epileptic seizures using raw scalp EEG signals. The proposed CNN-transformer deep learning architecture achieved a BA of 97.32%, enabling confident seizure predictions up to 76.8 min prior to seizure onset on average. By introducing the CIOPR metric, we illustrated the significant impact of preictal class definition on model behavior and demonstrated how these variations can be quantified to determine the OPP to maximize the model's usability for each individual patient.

Table 1. Overall results for selected preictal definition per patient. Notations: FAR = false alarm rate (computed based on the EPOCH [43] method), AUC = area under the receiver operating characteristic (ROC) curve, SPC = seizure prediction convergence, OPP = optimal preictal period.

Case	Sensitivity (%)	Specificity (%)	FAR (h^{-1})	Accuracy (%)	$F1$-score (%)	AUC (%)	SPC (mins)	OPP (mins)	Criterion
chb01	100.0	99.85	1.06	99.93	99.93	99.99	61.8	45	CIOPR
chb02	99.76	96.32	26.5	98.04	98.08	99.86	66.7	60	CIOPR
chb03	99.87	97.56	17.6	98.71	98.73	99.93	N/A	30	$F1$-score
chb04	99.84	99.19	5.83	99.52	99.40	99.99	175.5	45	CIOPR
chb05	99.62	96.66	24.1	98.14	98.16	99.87	54.9	60	CIOPR
chb06	94.76	73.54	190	84.15	85.71	92.90	52.9	60	CIOPR
chb07	99.54	97.74	16.2	98.64	98.66	99.72	61.5	60	CIOPR
chb09	99.72	98.85	8.30	99.28	99.29	99.98	93.5	60	CIOPR
chb10	98.40	92.84	51.6	95.62	95.76	99.54	68.8	60	CIOPR
chb11	99.86	99.81	1.37	99.83	99.83	99.99	N/A	45	$F1$-score
chb14	97.85	86.81	95.0	92.33	92.73	97.68	52.1	60	CIOPR
chb16	99.64	97.00	21.6	98.32	98.33	99.82	62.5	60	CIOPR
chb17	99.45	97.57	17.5	98.51	98.51	99.82	N/A	60	$F1$-score
chb18	99.59	96.40	25.9	98.00	98.02	99.43	56.3	60	CIOPR
chb19	100.0	99.64	2.60	99.82	99.82	100.0	N/A	30	$F1$-score
chb20	100.0	98.84	8.37	99.39	99.42	99.99	N/A	15	$F1$-score
chb21	99.74	97.03	21.4	98.37	98.41	99.82	139.6	45	CIOPR
chb22	99.19	86.38	98.1	92.79	93.23	99.38	50.0	45	CIOPR
chb23	100.0	99.36	4.60	99.68	99.68	99.99	N/A	60	$F1$-score
Mean	99.31	95.34	33.6	97.32	97.46	99.35	76.8		
± Std	1.20	6.38	46.0	3.76	3.41	1.61	36.8		

4.1. Comparison with literature

The BA metric (average of sensitivity and specificity) was used for comparison with existing literature to account for variations in the preictal-interictal data ratio across studies. Table 2 details the patient-level comparison with studies that report the sensitivity and specificity for at least 5 overlapping cases in the CHB-MIT dataset.

Overall, the BA achieved by our CNN-Transformer model is comparable with the state-of-the-art studies. Additionally, cases *chb06* and *chb14* exhibited on average the lowest BA across the included studies. The limited number of patients and the absence of detailed pathological data in the CHB-MIT dataset prevent further exploration of the effect of various epilepsy sub-types on seizure prediction performance. Furthermore, at the chosen decision threshold (0.5 in our study), specificity was consistently lower than sensitivity, a pattern appearing also in other studies [22–24, 46].

Only two studies investigated the impact of varying preictal intervals on model performance, with ranges from 15 to 120 min in [10] and 5 to 15 min in [47]. Both showed improved specificity with longer preictal intervals, yet their overall performance metrics–same as the ones outlined in section 2.6–changed subtly across different windows. Given these marginal gains, researchers often prioritize the preictal interval that maximizes the number of seizures available for model training, typically selecting 30 min before the onset [22, 24, 33, 44, 46, 48, 49].

4.2. CIOPR alongside conventional metrics

The newly introduced CIOPR metric was used to comprehensively assess model performance and compare the effect of different preictal state definitions. Analysis in section 3.5 demonstrated that integrative metrics like CIOPR can unravel pronounced performance differences among different preictal definitions that can strengthen or alter a hypothesis around the OPP.

While the $F1$-score and other conventional metrics remain useful and can guide design choices, comparing high-performing classifiers necessitates more sophisticated and sensitive measures to capture subtle behavioral differences. Figure 8 compares the best-performing preictal definition in our study (60 min) with the most commonly used definition in the literature (30 min) across sensitivity, $F1$-score, and CIOPR. Both intervals showed similar performance in terms of sensitivity, while the improvements in the overall $F1$-score using the 60 min one remained marginal, aligning with existing literature. When using CIOPR however, we were able to outline significant performance improvements associated with the 60 min window in terms of the hypothesized ideal model behavior (early prediction, low error, output consistency, and sharp transition).

Furthermore, the integrative nature of CIOPR enables a detailed breakdown of the individual measures comprising it to thoroughly explore specific output characteristics. Figure 9 illustrates how these measures were used to comprehensively assess the

Table 2. Balanced accuracy (%) comparison between state-of-the-art literature and the proposed solution. The table includes studies reporting both sensitivity and specificity measures for at least five cases of the CHB-MIT dataset overlapping with our work..

Case	Tsiouris *et al* [10]	Wang *et al* [44]	Xia *et al* [22]	Zhu *et al* [23]	Qi *et al* [33]	Zhang *et al* [45]	Ma *et al* [20]	Qi *et al* [46]	Ryu *et al* [47]	Jemal *et al* [48]	Yang *et al* [49]	Xu *et al* [32]	Average	Ours
chb01	100.00	99.00	100.00	99.46	99.56	98.95	99.99	97.81	100.00	96.75	99.65	98.35	99.13	99.93
chb02	100.00	99.00	97.93	99.47	94.70	93.15	89.48	96.12	86.94	95.55	75.75	N/A	93.46	98.04
chb03	100.00	100.00	99.17	96.02	96.49	97.51	97.74	88.25	96.82	98.80	95.90	92.20	96.57	98.71
chb04	99.97	99.27	N/A	N/A	N/A	97.07	N/A	N/A	78.26	91.35	N/A	N/A	93.18	99.52
chb05	99.83	99.67	98.88	99.91	98.51	97.48	97.11	90.38	94.33	85.00	93.50	86.40	95.08	96.20
chb06	99.34	99.00	99.26	98.13	N/A	97.75	88.89	N/A	94.20	84.75	N/A	64.80	91.79	84.15
chb07	99.88	100.00	97.50	92.04	N/A	99.40	83.97	N/A	100.00	93.60	N/A	N/A	95.80	98.64
chb09	100.00	99.85	99.79	97.58	95.59	98.49	N/A	91.80	99.83	87.75	79.05	N/A	94.97	99.28
chb10	99.91	97.83	99.86	99.64	93.70	93.68	93.37	90.42	90.53	90.70	80.85	N/A	93.68	95.62
chb11	100.00	99.94	N/A	96.39	N/A	98.16	99.65	N/A	100.00	99.30	N/A	89.30	97.84	99.83
chb14	99.86	100.00	98.22	99.65	99.65	86.63	89.93	90.56	89.67	74.70	69.30	79.60	89.81	92.33
chb16	99.75	99.42	99.17	96.72	N/A	92.63	N/A	N/A	81.03	89.75	N/A	N/A	94.06	98.32
chb17	99.95	99.76	98.61	97.98	N/A	95.02	N/A	N/A	100.00	97.05	N/A	N/A	98.34	98.51
chb18	99.66	99.87	N/A	99.89	97.88	88.89	N/A	94.33	92.35	96.10	96.10	N/A	96.12	98.00
chb19	100.00	98.65	N/A	100.00	98.25	97.30	98.83	95.97	100.00	99.50	95.80	N/A	98.43	99.82
chb20	100.00	98.89	100.00	99.99	99.79	99.72	99.95	97.01	100.00	98.90	99.35	97.85	99.29	99.39
chb21	100.00	98.98	98.34	96.88	99.78	89.40	95.55	97.85	96.91	89.10	92.70	78.65	94.51	98.37
chb22	100.00	97.36	N/A	97.19	N/A	94.96	93.19	N/A	91.46	88.35	N/A	82.75	93.16	92.79
chb23	100.00	95.00	99.38	99.97	99.80	93.13	99.75	99.52	94.64	94.75	98.90	98.85	97.81	99.68
Mean	99.90	99.02	99.01	98.16	97.81	95.23	94.81	94.17	94.05	92.20	89.74	86.88	95.42	97.22
± Std	0.16	1.19	0.77	2.03	2.08	3.71	4.94	3.57	6.38	6.23	10.08	10.27	2.56	3.76

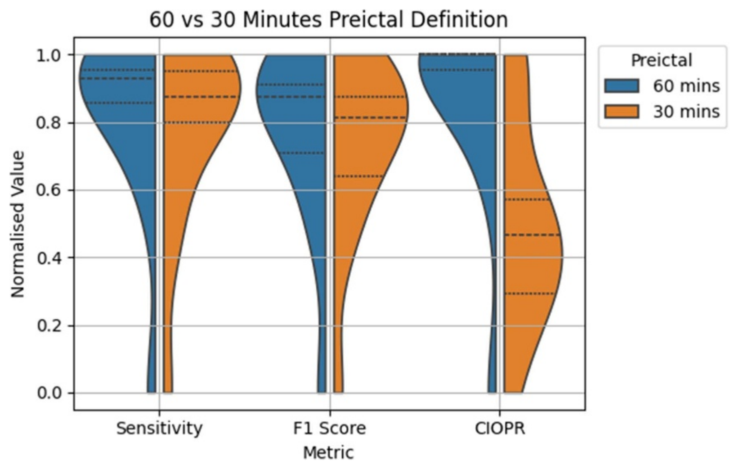
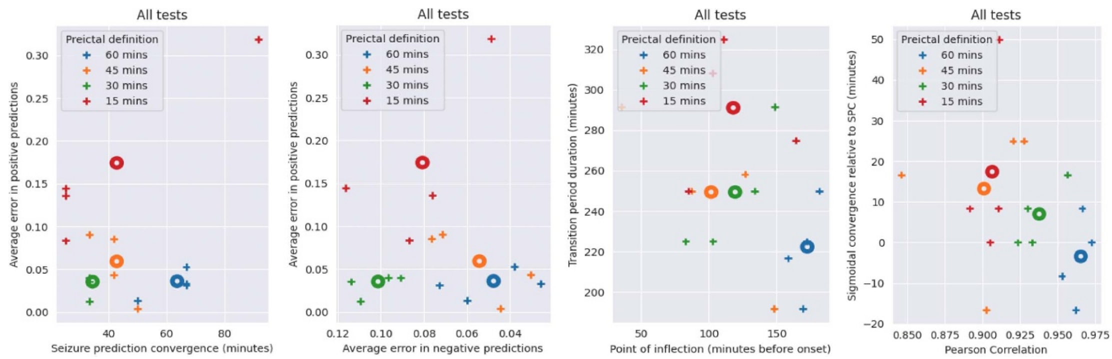


Figure 8. Distribution of normalized sensitivity, $F1$-score, and CIOPR values across eligible subjects for the 60 and 30 min preictal definitions.



(a) Average error in positive predictions versus SPC.

(b) Average error in positive versus negative predictions.

(c) Transition period duration versus point of inflection.

(d) Convergence points versus Pearson correlation.

Figure 9. Performance measures used for CIOPR calculation of seizure #1, case *chb16* across different preictal state definitions. The best performance corresponds to the bottom-right corner. Sub-figure 9(d) assesses the quality of the sigmoidal fitting, where the difference between the convergence points of the fitted curve and the model's output (seizure prediction convergence (SPC)) are shown in the vertical axis. Data points marked as 'plus' in the plots represent different training runs per preictal definition, while 'ring' markers correspond to the mean values across the runs.

performance for case *chb16*. The advantages of a 60 min preictal definition–early prediction time, minimized positive and negative errors, short TP, and alignment with the fitted sigmoidal curve representing ideal behavior–can be visualized in an interpretable manner that conventional metrics lack.

4.3. Preictal length definition and model behavior

Understanding how and why the chosen preictal period influences model behavior can prove critical in designing systems tailored to the patients' needs and addressing ongoing limitations in epileptic seizure prediction research. In particular, specificity increased with shorter preictal lengths, while sensitivity remained relatively invariant. This invariability combined with the consistently high scores could be suggestive of a key limitation in the field: insufficient data volume. Data oversampling in the preictal state along with lack of external validation could lead to model overfitting, where classifiers memorize rather than generalize training data [4]. Although the LOOCV approach prevents the model from encountering test seizures during training, the uniform data acquisition setting and the close temporal proximity of recordings might still result in high similarity between training and testing sets [50]. Conversely, interictal data, recorded throughout the whole day, exhibit greater variability from circadian rhythms and additional EEG patterns [51], complicating their classification. This greater variability might explain the consistently lower specificity and its significant reduction with shorter preictal durations, as these entail fewer training samples. Concordance with the literature results indicates that the aforementioned limitations are broadly applicable to the field and could inform future research directions.

Although specificity increased with longer preictal definitions, ND_{err} did not follow the same trends. This can be explained by ND_{err} being measured only within 10 h before onset, unlike specificity, which was calculated using all interictal state data. Thus, ND_{err} was computed on the subset of interictal data that was closer to the preictal-interictal boundary. Models trained with longer preictal intervals could detect preictal-like dynamics many hours before onset, leading to an earlier gradual increase in positive predictions and consequently earlier Infl. Hence, the increased ND_{err} could be related to the impending seizure. However, positive predictions this far from the onset could cause patient distress and be considered undesired in a clinical setting. On the other hand, longer preictal intervals led to reduced SPC_{err} values, earlier classifier convergence, and higher correlation coefficients with the fitted curve. This translates to higher accuracy, earlier prediction, and reduced output fluctuations, respectively, which could be desired in a clinical setting.

Extended preictal periods were expected to help classifiers learn spatiotemporal EEG features farther from the seizure onset, reducing the uncertainty during the interictal-to-preictal transition. TP values decreased with longer preictal lengths but showed no statistical significance. Interestingly, the lowest TP was achieved with the 45 min preictal length, suggesting that features learned farther than this point might often have diminished relevance for epileptic seizure prediction. Conversely, features learned within the last 30 min before onset achieve high prediction accuracy but may be insufficient for early identification of an impending seizure. This could lead to increased uncertainty during the TP and potentially confusion in a clinical implementation setting through prolonged output fluctuations. This is further highlighted by the fact that across all metrics among adjacent preictal definitions, the 45-to-30 min comparison yielded the highest number of significant differences.

4.4. Cross- and intra-patient heterogeneities

The CIOPR-related measures and their sensitivity to different preictal period lengths varied across testing seizures. This observation underscores the major challenge in epileptic seizure prediction research: inter-seizure heterogeneities, both across and within patients. These differences caused the OPP to fluctuate between 60 and 45 min across patients, and in some cases (e.g. *chb01*, *chb22*) within seizures of the same patient. Cross-patient variations can be addressed by designing patient-specific models. Intra-patient variations can be managed by choosing the best average across seizures, as in our study, or selecting a different preictal length definition for each training seizure, as suggested by [5].

These approaches, though effective at improving the classification of interictal and preictal segments, cannot eliminate inconsistent values across measures comprising CIOPR, due to the unique electrophysiological signature of each epileptic seizure. Among these measures, prediction time is of utmost importance as it directly influences alarm generation and provides a window for potential intervention. Keeping two variables among *Patient ID, Seizure ID, Preictal Length* fixed, and computing the mean standard deviation on the third showed that variations in prediction time could be primarily attributed to cross-patient heterogeneities (±38.3 min), followed by intra-patient heterogeneities (±17.7 min), and lastly, by the preictal period lengths used for training (±13.2 min).

The clinical usefulness of a seizure prediction system would rely on meeting the patients' preferences in prediction time and sensitivity-specificity trade-off [3]. Researchers in [52] highlighted that patient satisfaction was maximized with short prediction times, allowing effective lifestyle changes. Long prediction times on the other hand could be more useful in a closed-loop system, allowing room for drug-delivery or stimulation [53]. However, the observed cross-patient variations suggest that prediction time is not merely a matter of preference but is intrinsically linked to the preictal characteristics of each patient. Additionally, intra-patient variations, although less pronounced, could significantly impact the effectiveness of seizure predictions. For instance, in case *chb09*, the SPC ranged from 63.9 to 125 min.

A reliable system should predict seizures earlier than a minimum seizure prediction horizon (SPH) with fluctuations not exceeding the seizure occurrence period (SOP), such that prediction times range within the $(\mathrm{SPH}, \mathrm{SPH}+\mathrm{SOP})$ interval [54]. SPH should be large enough to allow clinical intervention (e.g. 30 min in [54]), while SOP could be limited to the duration of the treatment; e.g. 30 min for some anti-epileptic drugs [54]. Cross-patient heterogeneities could enable setting a realistic SPH interval, informed by both the patient's preferences and the electrophysiological signature of the preictal state. Case *chb04* for instance exhibited SPC values of 133.3 and 217.5 min before the onset. A 30 min SPH selection would hence prove non-practical for this case, as it would require a SOP of at least 3 h, leading to increased uncertainty for the patient. Similarly, understanding intra-patient heterogeneity patterns could allow providing realistic SOP intervals for each individual. For instance, at the OPP, case *chb07* experienced considerably less variations in prediction time (±4.7 min) than case *chb09* (±30.6 min).

4.5. Limitations and future work

Our study utilized raw scalp EEG signals with minimal pre-processing. Although the observed classification performance was comparable to state-of-the-art results, the 45 Hz low-pass cut-off could have excluded high-frequency information relevant for detecting preictal-state dynamics. The automatic selection of $N = 1690$ taps using the MNE package for the band-pass filter, though not an issue in our case, could significantly increase computational complexity in real-time systems. Additionally, limiting the postictal state to only 1 h could lead to contamination of interictal data with seizure-related effects and diminish the model's specificity. However, it provided a greater number of eligible seizures while maintaining competent classification performance.

Besides the risk of overfitting, performance results could be affected by data leakage, a prevalent issue in machine learning-related research [55]. In particular, the LOOCV approach might introduce temporal leakage, due to the inclusion of 'future' data (i.e. seizures) in the training process. For example, a model trained on seizures No. 1, 3, 4, 5 and tested on seizure No. 2 had access to information that occurred both before and after the test seizure, potentially introducing temporal dependencies between the training and testing sets. This information would not be available in a realistic implementation setting and could inflate the observed performance.

Furthermore, the classifiers' output behavior was not uniform across all subjects and seizures, introducing limitations in the generalizability of the CIOPR metric. Convergence did not always occur right before seizure onset (figure 4(b)), while in several cases, persistent output fluctuations complicated the identification of the SPC start (figure 4(d)). Additionally, the TP was directly computed from the fitted curve, making it dependent on the quality of the fitting. Figure 7 highlighted the issue, where small variations in model behavior caused large differences in the stretch of the sigmoidal curve and consequently in the calculated TP. Lastly, the penalty introduced due to long TPs was dependent on the interictal duration, ND, inhibiting direct comparisons across different seizures.

Future efforts should prioritize improving the quality of EEG datasets. Extended recording times and longitudinal acquisitions are necessary to assess algorithms' generalizability over time. Such a dataset could alleviate the need for the LOOCV approach and consequently mitigate data leakage issues. Reporting pathological information, such as the location of the seizure onset zone, clinical semiology, and epilepsy sub-type, would be crucial in drawing clinically relevant conclusions, including which seizure sub-types could be more challenging to predict. Additionally, it could enable an understanding of preictal state-related heterogeneities across seizures and patients. Unraveling those patterns could enable tailored identification of SPH and SOP intervals, that could be dynamically adjusted during the course of the day, depending on the individual and epilepsy pathology. This would ultimately lead to providing context-aware warnings and improving the clinical reliability of automated seizure prediction models.

The proposed solution would also benefit from developing a channel-selection algorithm at the input stage, as suggested by authors in [56]. Furthermore, implementing a post-processing scheme to filter out incorrect predictions and generate confident alarms would be useful. Clinically relevant metrics such as FAR (h^{-1}) should then be computed on the post-processed output and reported for a predefined SPH and SOP. While the CIOPR algorithm prioritizes prolonged prediction times, future improvements should tune it to emphasize convergence within the preferred $[\mathrm{SPH}, \mathrm{SPH} + \mathrm{SOP}]$ range. Consistent EEG datasets will enable benchmarking on achieved CIOPC values by fixing the input signal duration and the contribution of each term (e.g. TP).

5. Conclusions

We introduced a CNN-Transformer model for predicting epileptic seizures using scalp EEG signals and proposed the novel CIOPR metric. Using the CIOPR metric we established that varying preictal period lengths result in statistically significant differences in model behavior. Overall, increasing the preictal length led to earlier prediction times, sharper interictal-preictal transitions, and reduced output fluctuations in the preictal state, at the expense of an increased volume of positive predictions several hours before the onset. This finding highlights the need for careful selection of the OPP depending on the desired usability.

To the best of our knowledge, this is the first study that uses a fitting curve to model the output of a seizure prediction classifier. The newly introduced CIOPR metric provides interpretability on model performance that conventional metrics lack, by outlining how the distribution of positive predictions varies over time. With the increasing complexity and classification capabilities of deep learning architectures, more intuitive and sophisticated measures are required to capture subtle performance differences. Integrative measures like CIOPR, apart from being useful in selecting the most suitable deep learning model, they can provide meaningful clinical insights and shed light on future research directions.

Data availability statement

The raw CHB-MIT dataset that supports the findings of this study are available in the following links: https://doi.org/10.13 026/C2K01R; and further inquiries can be directed to the corresponding author.

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